DESIGN AND DEVELOPMENT OF NEXUS-AI

An AI based UNIVERSAL PROBLEM SOLVING SYSTEM(UPSS)

Team-DATAFORGE

Corner Stone Project Report

Milestone II

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(Declared under Distinct Category by Ministry of Education, Government of India)
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1. Objective

The objective of Milestone II is to develop the core intelligence and autonomous planning layer of Nexus-AI. This layer enables the system to understand a user's problem, identify its intent and domain, evaluate its complexity, and transform the problem into a structured, dependency-aware execution plan.

Instead of directly generating an answer, Nexus-AI first determines what needs to be done, in what order, and what tasks are required to solve the problem.

2. Core Problem Addressed

Complex real-world problems cannot always be solved through a single AI response. They may require multiple steps, different types of reasoning, external tools, specialized agents, and task dependencies.

Milestone II addresses this challenge by introducing an intelligent planning pipeline:

User Problem

↓

Understand the Problem

↓

Identify Intent

↓

Detect Domain

↓

Assess Complexity

↓

Decompose Problem

↓

Identify Dependencies

↓

Generate Execution Plan

The resulting plan becomes the foundation for the Agent, Tool, Provider, and Execution layers developed in subsequent milestones.

3. Implemented Components

A. Input Processing

The incoming user request is converted into a structured representation that can be processed by the intelligence pipeline.

Responsibilities include:

- Query normalization
- Input cleaning
- Structured problem representation
- Preparation of information for downstream reasoning

B. Intent Understanding

The system determines what the user is actually trying to accomplish.

It extracts relevant information such as:

- Primary objective
- Entities
- Constraints
- Requirements
- Preferences
- Expected outcome

For example:

User:

"Find the cheapest flight from Delhi to Mumbai

and compare it with available trains."

Intent:

Travel Search + Comparison

Entities:

Delhi → Mumbai

Constraints:

Cheapest option

Required Operations:

Flight Search

Train Search

Price Comparison

Recommendation

C. Domain Detection

Nexus-AI identifies the domain associated with the problem.

Examples include:

Travel

Finance

Coding

Research

Communication

Browser

Filesystem

Database

Calendar

Email

Documents

The architecture is designed to remain domain-agnostic, allowing additional domains to be introduced without redesigning the complete planning system.

D. Complexity Assessment

The system evaluates the complexity of a problem before deciding how it should be handled.

The assessment considers factors such as:

- Number of required tasks
- Task dependencies
- Required resources
- Reasoning requirements
- Potential need for multiple agents
- Potential execution complexity

The problem can then be categorized as:

Simple

Medium

Complex

This information helps determine the appropriate execution strategy in later stages.

4. Autonomous Planning

The Planner is the central component of Milestone II.

It converts the interpreted user problem into a structured execution plan.

For example:

User Request

↓

Planner

↓

Task 1 → Search Flights

↓

Task 2 → Search Trains

↓

Task 3 → Compare Results

↓

Task 4 → Generate Recommendation

When dependencies exist:

Task 1 ──────────────┐

├──→ Task 3 ──→ Task 4

Task 2 ──────────────┘

This allows later execution components to determine which tasks can run independently and which must wait for previous results.

5. Planner Task & Execution Plan

The planning layer represents individual operations using structured Planner Task objects.

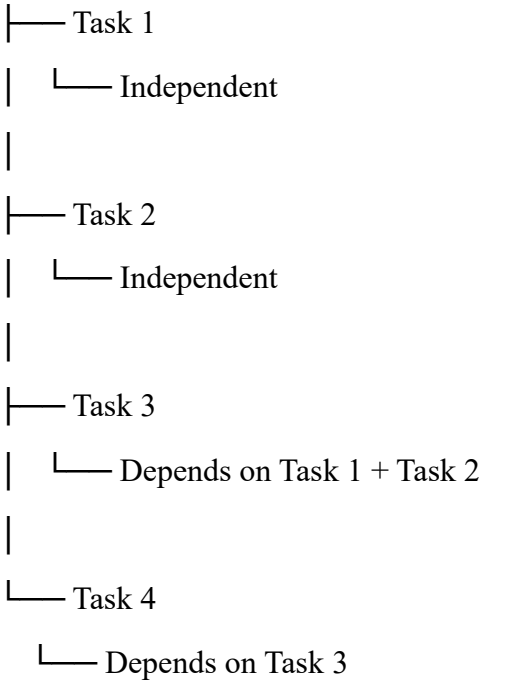
A task contains information required for downstream execution, such as:

- Task objective
- Required capabilities
- Input information
- Dependencies
- Execution requirements
- Expected outcome

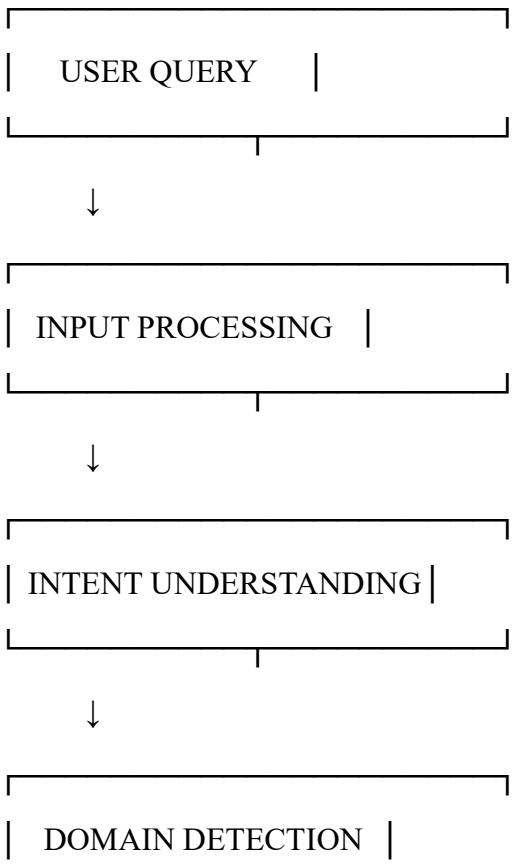
Multiple PlannerTask objects form an Execution Plan.

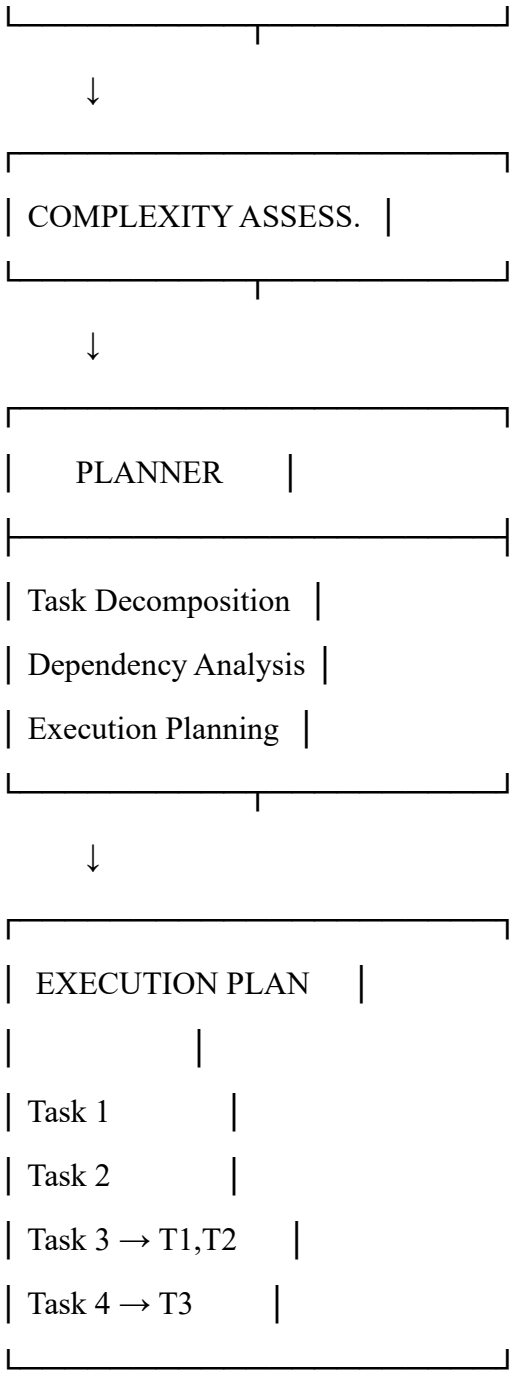
Execution Plan

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6. Milestone II Workflow





7. Example End-to-End Planning

User Query

"Research the best laptops under ₹80,000 for programming, compare their specifications and recommend the best option."

System Analysis

Intent:

Product Research + Comparison + Recommendation

Domain:

Research / Technology

Complexity:

Complex

Requirements:

- Find relevant laptops
- Collect specifications
- Compare specifications
- Evaluate programming suitability
- Recommend the best option

Generated Execution Plan

Task 1:

Find suitable laptops under ₹80,000

Task 2:

Collect specifications and relevant information

Task 3:

Compare shortlisted laptops

Depends on: Task 1, Task 2

Task 4:

Evaluate programming suitability

Depends on: Task 3

Task 5:

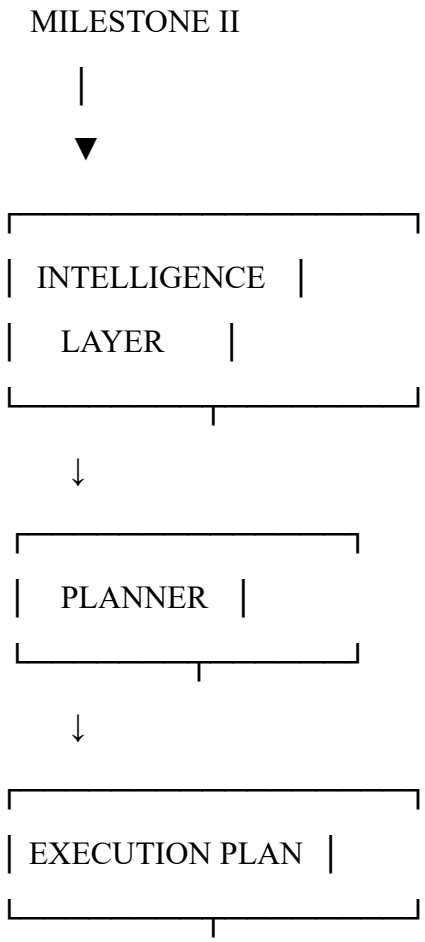
Generate final recommendation

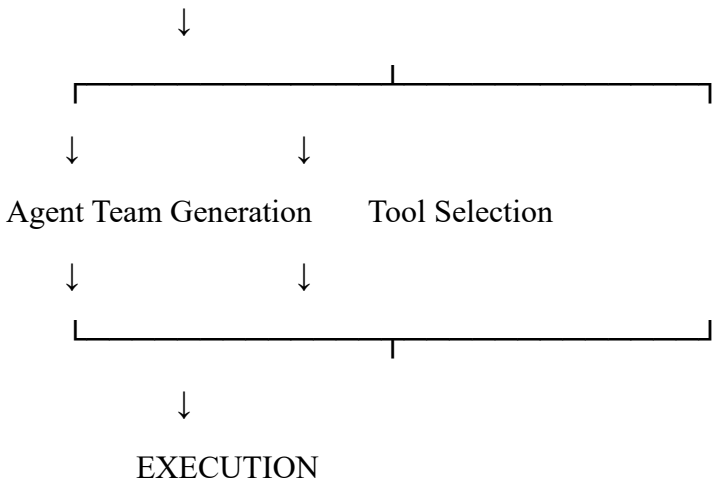
Depends on: Task 4

The execution plan can subsequently be passed to the Agent Team Generation and Tool Selection layers.

8. Architecture Position

Milestone II establishes the foundation for the remaining Nexus-AI system:





9. Key Outcomes

By completing Milestone II, Nexus-AI establishes the ability to:

- Understand complex user problems
- Extract intent and requirements
- Identify the relevant domain
- Assess problem complexity
- Break complex problems into smaller tasks
- Identify task dependencies
- Generate structured execution plans
- Provide a planning foundation for multi-agent execution
- Pass structured tasks to downstream system components